ARTIFICIAL INTELLIGENCE: CONCEPTS AND ECHNIQUES

WEEK 9:

Lecture 41: Multilayer Feed Forward Neural Network

This document, **"Lecture 41: Multilayer Feed Forward Neural Network,"** discusses Multilayer Perceptron Neural Networks (MLP NNs), their architecture, and applications.

Key points include:

● **Multilayer Perceptron Neural Network:**

○ No connections within a layer or direct connections between input and output layers.
○ Fully connected between layers, often with more than three layers.
○ The number of output units, and hidden units per layer, can differ from the number of input units.
○ Each unit functions as a perceptron.

● **Backpropagation:**

Details the mathematical calculations for activation, output, error, and weight updates between hidden and output units.

● **Applications:**

○ Digit Recognition (e.g., using 16 input values and 4 output bits to classify digits 0-9).
○ General applications include pattern classification, image processing, speech analysis, optimization problems, stock market forecasting, simulation, and robot steering.

● **Design Issues:**

Covers the number of nodes in input and output layers, network topology (number of hidden layers and nodes, feed-forward or recurrent), weight and bias initialization, and handling missing values in training examples.

● **Characteristics of ANNs:**

○ Multilayer NNs with at least one hidden layer are universal approximators.
○ They can handle redundant features by learning small weights for them.
○ NNs are sensitive to noise.
○ Training can be time-consuming, but classification of test examples is rapid once training is complete.
○ Only network information needs to be stored for testing, reducing space requirements.

Lecture 42

This file, **"lecture42.pdf,"** covers various topics related to hypothesis learning, class imbalance, and performance metrics in classification models.

Here's a summary of its key sections:

● **Hypothesis Learning:**

○ Uses logical inference to find a hypothesis that matches examples.
○ Hypotheses are logical sentences that classify examples.
○ The aim of inductive learning is to find a hypothesis that classifies well and generalizes to new examples.
○ Discusses how hypotheses inconsistent with new examples are removed.
○ Introduces "false negatives" (hypothesis says negative, actually positive) and "false positives" (hypothesis says positive, actually negative).
○ The "Current-Best-Learning" algorithm is presented, which searches for a consistent hypothesis and backtracks when needed.

● **Version Space Learning (Candidate Elimination Algorithm):**

○ An alternative to backtracking, it keeps all hypotheses consistent with the data so far.
○ Each new example removes inconsistent hypotheses, shrinking the "version space."
○ This approach is incremental, meaning old examples don't need re-examination.
○ The challenge of representing an enormous hypothesis space is addressed by "Least Commitment Search (LCS)."

● **Least Commitment Search (LCS):**

○ Uses a partial ordering (generalization/specialization) on the hypothesis space.
○ The entire version space is represented by two boundary sets: a "most general boundary" (G-set) and a "most specific boundary" (S-set).
○ All hypotheses between the S-set and G-set are consistent with the examples.
○ Describes how to initialize (G-set to True, S-set to False) and update S and G sets when new examples arrive, handling false positives and false negatives.
○ Discusses three possible outcomes: a single hypothesis remains, the version space collapses (no consistent hypothesis), or multiple hypotheses remain (representing a disjunction).

● **Drawbacks to LCS:**

○ Susceptible to collapsing if the domain contains noise or insufficient attributes.
○ Can lead to exponentially growing S-set or G-set elements in some hypothesis spaces.

● **Class Imbalance Problem:**

○ Occurs when one class (majority class) is significantly more represented than others (minority class).
○ Standard classifiers tend to be biased towards the majority class, leading to poor performance on the minority class.
○ Examples include fraud detection, anomaly detection, and medical diagnosis.

● **Approaches to Address Class Imbalance:**

○ **Data Level (Re-Sampling):**

■ Oversampling: Increases minority class instances (e.g., simple duplication, SMOTE). Can improve performance but risks overfitting. SMOTE creates synthetic samples.

■ Undersampling: Decreases majority class instances (e.g., random undersampling, Tomek Links). Can reduce training time but risks losing important data.

○ **Algorithmic Level:**

Cost-Sensitive Learning, Ensemble Methods, Focal Loss.

○ **Other Approaches:**

Batch Size Considerations, Feature Selection.

● **Key Considerations for Resampling:**

○ Severity of imbalance, dataset size, specific model, and risk of overfitting.

● **Performance Metrics:**

○ **Confusion Matrix:**

A table describing a classification model's performance, showing true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN).

○ **Precision:**

The ability to retrieve mostly relevant documents.

$$P = \frac{TP}{TP + FP}$$

○ **Recall:**

The ability to find all relevant items.

$$R = \frac{TP}{TP + FN}$$

○ **F-Measure (F1/Harmonic Mean):**

Combines precision and recall, useful for skewed data where accuracy is not appropriate.

$$F = \frac{2RP}{R + P}$$

○ **Accuracy:**

Overall correctness.

$$Accuracy = \frac{TP + TN}{TP + FP + FN + TN}$$

Emphasizes that accuracy alone is insufficient for imbalanced datasets.

○ **ROC Curves:**

Visualize a classifier's performance across different thresholds, illustrating the trade-off between True Positive Rate (TPR) and False Positive Rate (FPR). Higher curves indicate better performance.

● **Cost-Sensitive Learning:**

○ Aims to minimize the expected cost of misclassifications, as different errors can have different costs (e.g., medical diagnosis, fraud detection).
○ Illustrates with a "Cost Matrix" how a model with lower accuracy might be preferred if it minimizes overall cost.